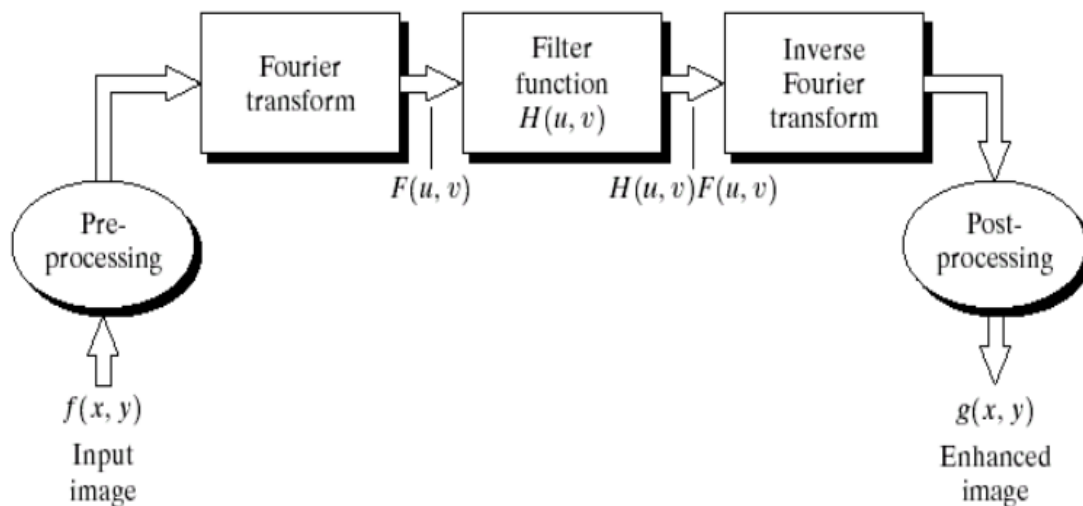

Unit 3

IMAGE ENHANCEMENT IN FREQUENCY DOMAIN

3.1. Introduction:

- In the frequency domain, a digital image is converted from spatial domain to frequency domain and image filtering is used for image enhancement for a specific application.
- There are two types of filters. They are High pass filters and low pass filters.
- For smoothing an image, low filter is implemented and for sharpening an image, high pass filter is implemented.
- The low pass filters and high pass filters have sub categorized i. e. ideal filter, Butterworth filter and Gaussian filter.
- The **ideal lowpass filter** is used to cut off all the high-frequency components of Fourier transformation.
- **Butterworth Lowpass Filter** is used to remove high-frequency noise with very minimal loss of signal components if order is high otherwise it reduces the noise and smoothens the transition if order is low.
- A **Gaussian low pass filter** is used for smoothens the transition and blurring regions of an image.
- Similarly,
- The **ideal high pass filter** is used to cut off all the low-frequency components of Fourier transformation.
- **Butterworth Lowpass Filter** is used to remove low-frequency noise with very minimal loss of signal components if order is high otherwise it reduces the noise and sharpens the edges if order is low.
- A **Gaussian low pass filter** is used for sharpening the edges

The image enhancement in frequency domain is shown below:



3.2. Fourier Transform:

- A Fourier transform is a tool to analyze and convert the image from spatial domain into frequency domain.
- The inverse Fourier transform brings back the image in spatial domain from frequency domain.
- Fourier transformation is a tool for image processing. it is used for decomposing an image into sine and cosine components.
- The input image is a spatial domain and the output is represented in the Fourier or frequency domain.
- Fourier transformation is used in a wide range of application such as image filtering, image compression. Image analysis and image reconstruction etc.
- **The formula for 1-D Fourier transformation and 1-D Inverse Fourier Transformation is given below:**

$$\begin{aligned} F(u) &= \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx, \\ f(x) &= \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du \end{aligned}$$

2D Fourier transform and 2D Inverse Fourier Transform:

$$\begin{aligned} F(u, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy, \\ f(x, y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv \end{aligned}$$

Properties of Fourier Transform:

1. **Duality:** It shows that if $h(t)$ possesses a Fourier transform $H(f)$, then the Fourier transform related to $H(t)$ is $H(-f)$.
2. **Linear transform:** Fourier transform comes under the category of linear transform. Let $g(t)$ and $h(t)$ be two Fourier transforms, which are represented by $G(f)$ and $H(f)$, respectively. In this scenario, it is easy to find the Fourier transform of the linear combination of h and g .
3. **Modulation property:** As per the modulation property, functions are modulated by the other functions if they are multiplied in time.

4. **Parseval's theorem:** As per this concept, the Fourier transform is unitary. This implies that the sum total of the squares of its Fourier transform, $H(f)$, equals the sum of squares of the function $h(t)$.

5. **Linearity:**

Addition of two functions corresponding to the addition of the two-frequency spectrum is called the linearity. If we multiply a function by a constant, the Fourier transform of the resultant function is multiplied by the same constant. The Fourier transform of sum of two or more functions is the sum of the Fourier transforms of the functions.

Case I.

If $h(x) \rightarrow H(f)$ then $ah(x) \rightarrow aH(f)$

Case II.

If $h(x) \rightarrow H(f)$ and $g(x) \rightarrow G(f)$ then $h(x)+g(x) \rightarrow H(f)+G(f)$

6. **Scaling:**

Scaling is the method that is used to change the range of the independent variables or features of data. If we stretch a function by the factor in the time domain then squeeze the Fourier transform by the same factor in the frequency domain.

$$\text{If } f(t) \rightarrow F(w) \text{ then } f(at) \rightarrow (1/|a|)F(w/a)$$

7. **Differentiation:**

Differentiating function with respect to time yields to the constant multiple of the initial function.

$$\text{If } f(t) \rightarrow F(w) \text{ then } f'(t) \rightarrow jwF(w)$$

8. **Convolution:**

It includes the multiplication of two functions. The Fourier transform of a convolution of two functions is the point-wise product of their respective Fourier transforms.

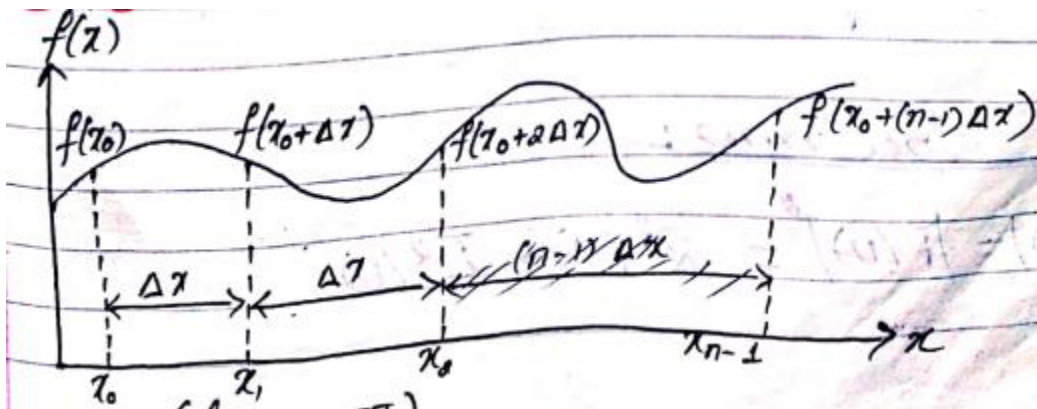
$$\text{If } f(t) \rightarrow F(w) \text{ and } g(t) \rightarrow G(w)$$

Then

$$f(t)*g(t) \rightarrow F(w)*G(w)$$

3.3. Discrete Fourier Transform:

- The Discrete Fourier Transform (DFT) is a specific kind of discrete transform, used in Fourier analysis.
- The DFT is a mathematical tool to convert continuous frequency into discrete one.
- It transforms time domain function into another function, which is called the frequency domain representation, or simply the DFT, of the original function.
- The DFT requires an input function that is discrete.
- Such inputs are often created by sampling a continuous function.
- The discrete input function must also have a limited (finite) duration, such as one period of a periodic sequence or a windowed segment of a longer sequence.



Here $f(x)$ is a continuous function to be discretized by taking 'n' samples i.e. $\{f(x_0), f(x_0 + \Delta x), \dots\}$

It is defined as:

The **1D Discrete Fourier Transform** of an image is given by:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}.$$

Where k is $0, 1, 2, \dots, N-1$

Here u is K

The **1D Inverse Discrete Fourier Transform** of an image is given by:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}.$$

Where n is $0, 1, 2, 3, \dots, N-1$

Here x is n according to our class lecturers.

The **2D discrete Fourier transform (DFT)** of an image f of size $M \times N$ is an image F of same size defined as:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

for $u = 0, 1, 2 \dots M-1$ and $v = 0, 1, 2 \dots N-1$.

- The **inverse 2D-DFT** is given by:

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

for $x = 0, 1, 2 \dots M-1$ and $y = 0, 1, 2 \dots N-1$

The Transform matrix of DFT is given by

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} 4 \times 4$$

To calculate the DFT using matrix method

For 1D image

$$\mathbf{F} = \mathbf{M} \cdot \mathbf{f}$$

For 2D Image

$$\mathbf{F} = \mathbf{M} \cdot \mathbf{f} \cdot \mathbf{M}^T \quad \text{Since the dft is symmetric the } \mathbf{M} \text{ and } \mathbf{M}^T \text{ are same}$$

Here $\mathbf{M}$ is the DFT transform matrix

(1) Fourier spectrum:

$$|F(u)| = \sqrt{R^2(u) + I^2(u)}$$

(2) Power spectrum:

$$|P(u)| = |F(u)|^2 = R^2(u) + I^2(u)$$

(3) Phase Angle (spectrum):

$$\phi = \tan^{-1} \left[\frac{I(u)}{R(u)} \right]$$

Where,

'u' is the transform frequency variable.
'x' is spatial or Image variable.

Properties:

- ❖ Linearity
- ❖ Periodicity
- ❖ Circular Time Shift
- ❖ Time Reversal
- ❖ Conjugation
- ❖ Circular frequency Shift
- ❖ Multiplication
- ❖ Circular Convolution
- ❖ Circular Correlation
- ❖ Parseval's Theorem

LINEARITY

Statement: The linearity property of DFT states that the DFT of a linear weighted combination of two or more signals is equal to similar linear weighted combination of the DFT of individual signals.

Let $\text{DFT}\{x_1(n)\}=X_1(K)$ & $\text{DFT}\{x_2(n)\}=X_2(K)$ then,

$$\text{DFT}\{a_1x_1(n)+a_2x_2(n)\}=a_1X_1(K)+a_2X_2(K)$$

Where a_1 & a_2 are constants.

PERIODICITY

Statement: If a sequence $x(n)$ is periodic with periodicity of N samples then N -point DFT, $X(K)$ is also periodic with a periodicity of N Samples

Let $x(n)=x(n+N)$ for all n

$$X(K)=X(K+N) \text{ for all } K$$

CIRCULAR TIME SHIFT

Statement: The Circular time shift property of DFT says that if a discrete time signal is circularly shifted in the time by ' m ' units then its DFT is multiplied by $e^{-j2\pi km/N}$

Let $\text{DFT}\{x(n)\}=X(K)$ then,

$$X(K) \cdot e^{-j2\pi km/N} = \text{DFT}\{x(n-m)_N\}$$

CONJUGATION

Statement: Let $x(n)$ be a complex N -point discrete sequence $x^*(n)$ be its conjugate sequence

If $\text{DFT}\{x(n)\}=X(K)$ then,
 $\text{DFT}\{x^*(n)\}=X^*(N-K)$

MULTIPLICATION

Statement: The Multiplication Property of DFT says that the DFT of product of two discrete time sequences is equivalent to circular convolution of DFT's of the individual sequences scaled by the factor of $1/N$

If $\text{DFT}\{x(n)\}=X(K)$ then,
 $\text{DFT}\{x_1(n) x_2(n)\}=1/N\{X_1(K) \odot X_2(K)\}$

PARSEVALS THEOREM

Statement: Let $\text{DFT}\{x_1(n)\}=X_1(K)$ & $\text{DFT}\{x_2(n)\}=X_2(K)$ then by Parseval's theorem

$$\sum_{n=0}^{N-1} x_1(n) x_2^*(n) = 1/N \sum_{K=0}^{N-1} X_1(K) X_2^*(K)$$

CIRCULAR CORRELATION

Statement: The Circular Correlation of two sequence $x(n)$ and $y(n)$ is defined as

$$r_{xy}(m) = \sum_{n=0}^{N-1} x(n) y^*((n-m))_N$$

Let $\text{DFT}\{x(n)\}=X(K)$ and $\text{DFT}\{y(n)\}=Y(K)$ then by Correlation property,

$$X(K) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \text{---(1)}$$

$$Y(K) = \sum_{p=0}^{N-1} y(p) e^{-j2\pi kp/N} \text{---(2)}$$

Consider the product of $X(K) Y^*(K)$. The IDFT of the product is given by,

$$\text{DFT}^{-1}\{X(K) Y^*(K)\} = 1/N \sum_{K=0}^{N-1} X(K) Y^*(K) e^{j2\pi km/N}$$

Let $n=m$

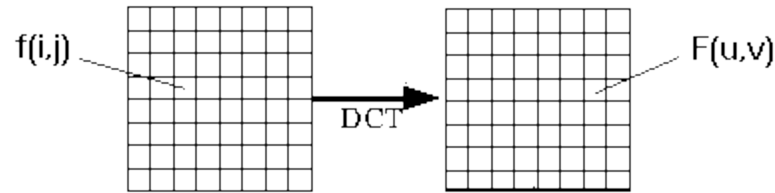
$$= 1/N \sum_{K=0}^{N-1} X(K) Y^*(K) e^{j2\pi km/N} \text{---(3)}$$

Applications:

- Signal analysis
- Sound filtering
- Data compression
- Cross correlation
- Matched filtering
- System identification
- Power spectrum estimation
- Display signal and spectrum

3.4. Discrete Cosine Transform:

- The DCT is similar to the discrete Fourier transform: it transforms a signal or image from the spatial domain to the frequency domain
- DCT convert an image into its equivalent frequency domain by partitioning image pixel matrix into blocks of size $N \times N$, N depends upon the type of image.
- For example, if we used a black & white image of 8 bit then all shading of black & white color can be expressed into 8 bits hence, we use $N=8$, similarly for color image of 24 bit we can use $N=24$ but using block size $N=24$ -time complexity may increase hence we operate DCT on individual color component for a color image. Color image consist of 8 bit red + 8 bit green + 8 bit blue hence we apply DCT on each color component (Red, Green, Blue) using block size $N=8$.



One-Dimensional DCT If we have one-D sequence of signal value of length N then its equivalent 1D-DCT can be expressed as

$$C(u) = \alpha(u) \sum_{x=0}^{N-1} f(x) \cos \left[\frac{\pi(2x+1)u}{2N} \right]$$

for $u = 0, 1, 2, \dots, N-1$.

1D-IDCT can be expressed as

$$f(x) = \sum_{u=0}^{N-1} \alpha(u) c(u) \cos \left[\frac{\pi(2x+1)u}{2N} \right]$$

Where $f(x)$ is signal value at point x
transform coefficient for value u .

$$\alpha(u) = \begin{cases} \sqrt{\frac{1}{N}} & \text{for } u = 0 \\ \sqrt{\frac{2}{N}} & \text{for } u \neq 0 \end{cases}$$

The **two-dimensional DCT** of an M -by- N matrix A is defined as follows:

$$B_{pq} = \alpha_p \alpha_q \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} A_{mn} \cos \frac{\pi(2m+1)p}{2M} \cos \frac{\pi(2n+1)q}{2N}, \quad 0 \leq p \leq M-1, \quad 0 \leq q \leq N-1$$

$$\alpha_p = \begin{cases} 1/\sqrt{M}, & p = 0 \\ \sqrt{2/M}, & 1 \leq p \leq M-1 \end{cases} \quad \alpha_q = \begin{cases} 1/\sqrt{N}, & q = 0 \\ \sqrt{2/N}, & 1 \leq q \leq N-1 \end{cases}$$

The **two-dimensional IDCT** of an M -by- N matrix A is defined as follows:

$$A_{mn} = \sum_{p=0}^{M-1} \sum_{q=0}^{N-1} \alpha_p \alpha_q B_{pq} \cos \frac{\pi(2m+1)p}{2M} \cos \frac{\pi(2n+1)q}{2N}, \quad 0 \leq m \leq M-1, \quad 0 \leq n \leq N-1$$

$$\alpha_p = \begin{cases} 1/\sqrt{M}, & p=0 \\ \sqrt{2/M}, & 1 \leq p \leq M-1 \end{cases} \quad \alpha_q = \begin{cases} 1/\sqrt{N}, & q=0 \\ \sqrt{2/N}, & 1 \leq q \leq N-1 \end{cases}$$

The DCT Transform Matrix

$$T_{pq} = \begin{cases} \frac{1}{\sqrt{M}} & p=0, \quad 0 \leq q \leq M-1 \\ \sqrt{\frac{2}{M}} \cos \frac{\pi(2q+1)p}{2M} & 1 \leq p \leq M-1, \quad 0 \leq q \leq M-1 \end{cases}$$

The 4x4 DCT transform matrix is given by

$$\begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.653 & 0.2705 & -0.2705 & -0.653 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.2705 & -0.653 & 0.653 & -0.2705 \end{bmatrix} \quad 4 \times 4$$

To calculate the DCT using matrix method

For 1D image

$$\mathbf{F} = \mathbf{C} \cdot \mathbf{f}$$

For 2D Image

$\mathbf{F} = \mathbf{C} \cdot \mathbf{f}$, $\mathbf{C}^T$ Since the DCT is not symmetric the C and $\mathbf{C}^T$ are not same. so we have to calculate the transpose of the C

Here C is the DCT transform matrix

Applications of DCT:

- DCTs are widely used in encoding, decoding, video, audio, multiplexing and analog to digital conversion.
- DCTs are also commonly used for High-definition Television (HDTV) encoder/decoder chips.
- DCT is widely used in JPEG compression as well.

Properties of DCT:

- De correlation
- Energy Compaction
- Separability
- Symmetry
- Orthogonality

Energy Compaction:

- Efficiency of a transformation scheme can be directly gauged by its ability to pack input data into as few coefficients as possible. This allows the quantizer to discard coefficients with relatively small amplitudes without introducing visual distortion in the reconstructed image. DCT exhibits excellent energy compaction for highly correlated images.

Separability:

- The principle advantage that $C(u, v)$ can be computed in two steps by successive 1-D operations on rows and columns of an image.

$$C(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos\left[\frac{\pi(2x+1)u}{2N}\right] \cos\left[\frac{\pi(2y+1)v}{2N}\right]$$

Symmetry:

- A separable and symmetric transform can be expressed in the form

$$T = AfA ,$$

where A is an $N \times N$ symmetric transformation matrix with entries

$a(i, j)$ given by,

$$a(i, j) = \alpha(j) \sum_{k=0}^{N-1} \cos \left[\frac{\pi(2j+1)k}{2N} \right] ,$$

and f is the $N \times N$ image matrix.

- This is an extremely useful property since it implies that the transformation matrix can be pre computed offline and then applied to the image thereby providing orders of magnitude improvement in computation efficiency.

Orthogonality:

- The inverse transformation as

$$f = A^{-1} T A^{-1} ,$$

DCT basis functions are orthogonal. The inverse transformation matrix of A is equal to its transpose i.e.

$$A^{-1} = A^T .$$

Therefore, and in addition to its de correlation characteristics, this property renders some reduction in the pre-computation complexity

3.5. Hadamard Transform:

- Hadamard transform is a non-sinusoidal, orthogonal transformation technique that decomposes a signal into a set of basic functions.
- These basis functions are Hadamard functions, which are rectangular or square waves with values of +1 or -1.
- The kernel of the Hadamard matrix is defined as:

⇒ The 1-D, forward Hadamard Kernel is given by the relation

$$g(x, u) = \frac{1}{N} (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

i.e. $g(x, u) = \frac{1}{N} (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$

⇒ The 1-D Hadamard transform is given by

$$H(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

where $N=2^n$, and u has values in the range $0, 1, 2, \dots, N-1$

⇒ The Hadamard Kernel forms a matrix having orthogonal rows and columns

⇒ The inverse Hadamard Kernel is given by

$$h(x, u) = (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

⇒ The inverse Hadamard Transform

$$f(x) = \sum_{u=0}^{N-1} H(u) (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

for $x=0, 1, 2, \dots, N-1$

The 2-D kernels are given by

$$g(x, y, u, v) = \frac{1}{N} (-1)^{\sum_{i=0}^{N-1} [b_i(x)b_i(u) + b_i(y)b_i(v)]}$$

and

$$h(x, y, u, v) = \frac{1}{N} (-1)^{\sum_{i=0}^{N-1} [b_i(x)b_i(u) + b_i(y)b_i(v)]}$$

⇒ The 2-D Hadamard transform pair

$$H(u, v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) (-1)^{\sum_{i=0}^{N-1} [b_i(x)b_i(u) + b_i(y)b_i(v)]}$$

and

$$f(x, y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} H(u, v) (-1)^{\sum_{i=0}^{N-1} [b_i(x)b_i(u) + b_i(y)b_i(v)]}$$

⇒ The Hadamard matrix of lowest order ($N=2$) is

$$H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

⇒ Letting H_N represent the matrix of order N , the recursive relationship is

$$H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$$

where H_{2N} is the Hadamard matrix of order $2N$ and $N=2^n$ is assumed

The transformation matrix is given by

$$A = \frac{1}{\sqrt{N}} H_N$$

$$H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

and

$$H_8 = \begin{bmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{bmatrix}$$

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

- (1) It is real, symmetric and orthogonal
ie
$$H = H^* \Rightarrow H^{-1} = H^T$$
- (2) It is a fast transform and 1-D Hadamard Transform can be implemented in $O(N \log_2 N)$ addition and subtraction.
- (3) It has only binary values ie 1 or -1 in its kernel matrix. No multiplications are required in the transform.
- (4) It is used for digital image processing and digital signal processing.
- (5) It has good energy compaction for highly co-related image.

Applications:

- Hadamard are used encoding, decoding, video, audio, multiplexing and analog to digital conversion.
- Hadamard transform are used for High-definition Television (HDTV) encoder/decoder chips.
- Hadamard is also used in JPEG and MPEG compression as well.

To calculate the Hadamard transform of the given image, following formula are needed:

For 1D image:

$$g = H \times f$$

For inverse

$$F = H^T \times g$$

For 2D image

$$g = H \times f \times H^T$$

For inverse

$F = H^T \times g \times H$, where H is Hadamard matrix and here H^T and H are same since symmetric matrix

3.6. Haar Transform:

HAAR TRANSFORM

It is a transform which is based on the class of orthogonal, non-sinusoidal, non-square functions whose elements are the multiples of $-1, +1, 0$ by the powers of $\sqrt{2}$.

It is a computational efficient transfer function for 'N' point whose vectors require $2(N-1)$ additions and N multiplications.

STEP 1: Now determine the value of " N ", which represents the order of the system or the size of the kernel.

STEP 2: Determine the total no of Bits required based upon the order of the function using the expression $n = \log_2 N$.

STEP 3: Determine the Values of the two specific terms p and q .

- (i) $p \in [0, n-1]$ i.e. $0 \leq p \leq n-1$
- (ii) If $p = 0$ then the value of q can be either '0' or '1'
- (iii) If $p \neq 0$ then $q \in [1, 2^p]$ i.e. $1 \leq q \leq 2^p$

HAAR TRANSFORM

STEP 4: Now determine the value of k , which tells us the total number of rows in the kernel by the equation $k = 2^p + q - 1$

STEP 5: Determine the value of the Z where $Z \in \left[0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}\right]$

STEP 6: If $k = 0$ then $H(Z) = \frac{1}{\sqrt{N}} \forall Z$

$$\text{else } H_k(Z) = H_{pq}(Z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{p/2} : \frac{q-1}{2^p} \leq Z < \frac{q-1/2}{2^p} \\ -2^{p/2} : \frac{q-1/2}{2^p} \leq Z < \frac{q}{2^p} \\ 0 : \text{otherwise} \end{cases}$$

STEP 7: When $k = 1$

$$H_k(Z) = H_{pq}(Z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{p/2} : \frac{q-1}{2^p} \leq Z < \frac{q-1/2}{2^p} \\ -2^{p/2} : \frac{q-1/2}{2^p} \leq Z < \frac{q}{2^p} \\ 0 : \text{otherwise} \end{cases}$$

now substitute the values of p and q in the above equation then we get

$$H_1(Z) = H_{01}(Z) = \frac{1}{\sqrt{2}} \begin{cases} +2^{0/2} : \frac{1-1}{2^0} \leq Z < \frac{1-1/2}{2^0} \\ -2^{0/2} : \frac{1-1/2}{2^0} \leq Z < \frac{1}{2^0} \\ 0 : \text{otherwise} \end{cases} \quad H_1(Z) = H_{01}(Z) = \frac{1}{\sqrt{2}} \begin{cases} +1 : 0 \leq Z < \frac{1}{2} \\ -1 : \frac{1}{2} \leq Z < 1 \\ 0 : \text{otherwise} \end{cases}$$

$\therefore H_1(0) = \frac{1}{\sqrt{2}}(1) = \frac{1}{\sqrt{2}}$
 $H_1(\frac{1}{2}) = \frac{1}{\sqrt{2}}(-1) = -\frac{1}{\sqrt{2}}$

STEP 8: Now the kernel for the haar Transform is formed by the values obtained **STEP 6** and **STEP 7** by $k = 0$ and $k = 1$.

	$z=0$	$z=1/2$
$k=0$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
$k=1$	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Algorithm of Haar transform kernel for N=4

DESIGN OF KERNEL FOR HAAR TRANSFORM of ORDER N=4

STEP 1: Here $N=4$ which represents the order of the system or the size of the kernel.

STEP 2: to determine the total no of Bits required based upon the order of the function using the expression $n = \log_2 N = \log_2 4 = 2 \log_2 2 = 1$.

STEP 3: to determine the Values of the two specific terms p and q .

(i) $p \in [0, n-1]$ i.e. $0 \leq p \leq n-1$ i.e.
 $p \in [0, 2-1]$ $0 \leq p \leq 2-1$
 $p = 0, 1$

(ii) If $p = 0$ then the value of q can be either '0' or '1'

(iii) If $p \neq 0$ then $q \in [1, 2^p]$ i.e. $1 \leq q \leq 2^p$ i.e. $1 \leq q \leq 2^1$ i.e. $1 \leq q \leq 2$ therefore $q = 1, 2$

STEP 4: Now let us determine the value of k which tells us the total number of rows in the kernel by the equation $k = 2^p + q - 1$ i.e.

if $p = 0$ and $q = 0$ then $k = 2^0 + 0 - 1 = 1 + 0 - 1 = 0$ that is $k = 0$

if $p = 0$ and $q = 1$ then $k = 2^0 + 1 - 1 = 1 + 1 - 1 = 1$ that is $k = 1$

if $p = 1$ and $q = 1$ then $k = 2^1 + 1 - 1 = 2 + 1 - 1 = 2$ that is $k = 2$

if $p = 1$ and $q = 2$ then $k = 2^1 + 2 - 1 = 2 + 2 - 1 = 3$ that is $k = 3$

therefore,

p	q	k
0	0	0
0	1	1
1	1	2
1	2	3

STEP 5: Based on N we need to determine the value of the Z where $Z \in \left[0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}\right]$

STEP 6: If $k = 0$ then $H(Z) = \frac{1}{\sqrt{N}} \forall Z$ i.e. $H(Z) = \frac{1}{\sqrt{4}} \forall Z$ i.e. $H(Z) = \frac{1}{\sqrt{4}} \forall Z \in \left[0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}\right]$

STEP 7: When $k = 1$ $p = 0$ and $q = 1$

$$H_k(Z) = H_{pq}(Z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{p/2} : \frac{q-1}{2^p} \leq Z < \frac{q-1/2}{2^p} \\ -2^{p/2} : \frac{q-1/2}{2^p} \leq Z < \frac{q}{2^p} \\ 0 : \text{otherwise} \end{cases}$$

now substitute the values of p and q in the above equation then we get

$$H_1(Z) = \frac{1}{\sqrt{4}} \begin{cases} +2^{0/2} : \frac{1-1}{2^0} \leq Z < \frac{1-1/2}{2^0} \\ -2^{0/2} : \frac{1-1/2}{2^0} \leq Z < \frac{1}{2^0} \\ 0 : \text{otherwise} \end{cases} \quad H_1(Z) = \frac{1}{\sqrt{4}} \begin{cases} +1 : 0 \leq Z < \frac{1}{2} \\ -1 : \frac{1}{2} \leq Z < 1 \\ 0 : \text{otherwise} \end{cases}$$

$$\begin{aligned} \therefore H_1(0) &= \frac{1}{\sqrt{4}}(1) = \frac{1}{\sqrt{4}} \\ \therefore H_1\left(\frac{1}{4}\right) &= \frac{1}{\sqrt{4}}(1) = \frac{1}{\sqrt{4}} \\ H_1\left(\frac{2}{4}\right) &= \frac{1}{\sqrt{4}}(-1) = -\frac{1}{\sqrt{4}} \\ H_1\left(\frac{3}{4}\right) &= \frac{1}{\sqrt{4}}(-1) = -\frac{1}{\sqrt{4}} \end{aligned}$$

STEP 8: Now the kernel for the haar Transform is formed by the values obtained **STEP 6** and **STEP 7** by $k = 0$ and $k = 1$.

	$z=0$	$z=1/4$	$z=2/4$	$z=3/4$
$k=0$	$\frac{1}{\sqrt{4}}$	$\frac{1}{\sqrt{4}}$	$\frac{1}{\sqrt{4}}$	$\frac{1}{\sqrt{4}}$
$k=1$	$\frac{1}{\sqrt{4}}$	$\frac{1}{\sqrt{4}}$	$-\frac{1}{\sqrt{4}}$	$-\frac{1}{\sqrt{4}}$
$k=2$	$\frac{\sqrt{2}}{\sqrt{4}}$	$-\frac{\sqrt{2}}{\sqrt{4}}$	0	0
$k=3$	0	0	$\frac{\sqrt{2}}{\sqrt{4}}$	$-\frac{\sqrt{2}}{\sqrt{4}}$

$$H = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$$

Properties of Haar Transform:

- Haar transform is real and orthogonal.
- It is very fast.
- It is separable and symmetric.
- The basis vectors are sequency ordered.
- It has poor energy compaction for images.

Applications of Haar Transform:

- JPEG compression
- MPEG compression
- Television broadcasting
- Encoding
- Decoding

To calculate the Haar transform of the given image, following formula are needed:

For 1D image:

$$\mathbf{g} = \mathbf{H} \times \mathbf{f}$$

For inverse

$$\mathbf{F} = \mathbf{H}^T \times \mathbf{g}$$

For 2D image

$$\mathbf{g} = \mathbf{H} \times \mathbf{f} \times \mathbf{H}^T$$

For inverse

$$\mathbf{F} = \mathbf{H}^T \times \mathbf{g} \times \mathbf{H}$$

NOTE: Since Haar matrix is not symmetric so $\mathbf{H}^T$ and $\mathbf{H}$ are not same